



A novel integrated method for improving the forecasting accuracy of crude oil: ESMD-CFastICA-BiLSTM-Attention

Zisheng Ouyang^a, Min Lu^a, Zhongzhe Ouyang^{b,*}, Xuewei Zhou^a, Ren Wang^c

^a Hunan Key Laboratory of Macroeconomic Big Data Mining and its Application, School of Business, Hunan Normal University, Changsha, 410081, PR China

^b Department of Biostatistics, University of Michigan, Ann Arbor, MI, 48109, USA

^c The School of Finance, Hunan University of Technology And Business, Changsha, 410205, PR China

ARTICLE INFO

JEL classification:

C45
C53
C55
G15
G17
Q47

Keywords:

Crude oil price and volatility
Extreme-point symmetric empirical mode decomposition
Fast independent component analysis
Deep learning

ABSTRACT

The volatility of crude oil price can significantly impact the stability of the crude oil market and even the global economy. Effectively predicting global crude oil price and volatility provides a scientific basis for decision-making for market regulators and investors worldwide, promoting the sound development of the global economy. In this study, we integrate signal processing with deep learning methods to present an optimal forecasting strategy for global crude oil price and volatility. We select the daily price of the WTI crude oil market from April 4, 1983, to December 12, 2023, for calculating volatility. Subsequently, employing extreme-point symmetric empirical mode decomposition (ESMD), K-means clustering, and fast independent component analysis method, we decompose and reconstruct the forecasting data, obtaining independent components with non-Gaussian characteristics. These components serve as inputs to estimate the accuracy of various models, including BiLSTM, Attention, LSTM, SVR, RF, and their combinations, in predicting crude oil price and volatility from both a point prediction and interval prediction perspective. Empirical results demonstrate that data decomposition, reconstruction, and the BiLSTM-Attention model outperform other models in predicting crude oil price and volatility.

1. Introduction

As an essential raw material and principal energy source for industrial activity, crude oil exerts an extremely significant role in global economic activities (Yang et al., 2021; Zhang et al., 2023a). However, against the current backdrop of the post-pandemic era and a plethora of major international geopolitical events such as Russia–Ukraine conflict, the Israel–Palestine conflict, and armed conflicts in Sudan, the global crude oil price is experiencing high-level volatility and becoming a major source of risk transfer. Numerous studies indicate that the volatility of the crude oil price, often characterized as nonlinear and non-static, has adverse effects on investments (Ilyas et al., 2021; Hasan et al., 2022), financial markets (Jiang et al., 2020; Luo et al., 2024) and may even lead to economic recessions (Kilian and Vigfusson, 2013; Badeeb et al., 2021). Consequently, establishing a stable and accurate model to predict the trend in crude oil market from various perspectives such as oil price and volatility is necessary to promote the stable operation of the national economy and protect policy security (Abdollahi and Ebrahimi, 2020; Yu et al., 2022).

- Considerable efforts in existing studies about crude oil price prediction have been devoted to constructing forecasting models, which

can be broadly categorized into three types: traditional econometric models, standalone machine learning models, and hybrid models (Guo et al., 2023; Zhang et al., 2024). Hybrid models can further be divided into combinations of data preprocessing methods with machine learning models and combinations of individual machine learning models. Traditional econometric forecasting models mainly involve moving average (MA), exponential smoothing (ES), vector autoregressive model (VAR), autoregressive integrated moving average (ARIMA), conditional heteroskedasticity model (ARCH), and autoregressive conditional heteroskedasticity model (GARCH) (Cheong, 2009; Guha and Bandyopadhyay, 2016; Wang et al., 2016; Jiang et al., 2020; Drachal, 2021; Wen et al., 2022). Nevertheless, these models have limitations in capturing non-linear features in the data. Due to the ability of artificial intelligence models to effectively overcome these limitations, numerous studies have made attempts to explore this direction (Jianwei et al., 2023).

Compared to traditional econometric models, machine learning forecasting models such as support vector machine (SVM) and decision tree (DT), can effectively enhance the forecasting accuracy by further

* Corresponding author.

E-mail addresses: ouyang_zs@hunnu.edu.cn (Z. Ouyang), ml2020@hunnu.edu.cn (M. Lu), oyzz@umich.edu (Z. Ouyang), xueweizhou@hunnu.edu.cn (X. Zhou), renwang@hutb.edu.cn (R. Wang).

<https://doi.org/10.1016/j.eneeco.2024.107851>

Received 4 February 2024; Received in revised form 5 August 2024; Accepted 13 August 2024

Available online 22 August 2024

0140-9883/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

mining the information within the sample (Bucci, 2020; Niu et al., 2023b,a; Li and Li, 2023). However, as the sample size expands, the limitations of machine learning become increasingly evident. To enhance accuracy, deep learning methods have been introduced into the forecasting of crude oil price. For instance, Chiroma et al. (2015) discovered the accuracy and efficiency of neural networks in crude oil price prediction. Moreover, recurrent neural network (RNN), especially long short-term memory network (LSTM) and gated recurrent network (GRU), have become crucial methods in addressing time series problems. And this is attributed to their ability to effectively capture sequential information (Busari and Lim, 2021; Urolagin et al., 2021; Ouyang et al., 2022). At the same time, convolutional neural network (CNN), bidirectional long short-term memory network (BiLSTM), and other deep learning models are widely applied in the prediction of time series data, including crude oil price (Chen and Huang, 2021; Ma et al., 2021; Ouyang et al., 2023).

In fact, crude oil price and volatility typically both exhibit characteristics such as a high degree of uncertainty, nonlinearity, and fluctuation. As part of time series forecasting, these traits make them challenging for a single model, whether it be a machine learning or deep learning model, to fully capture their changing features (Zhang et al., 2023b). Hence, most current studies are dedicated to constructing high-precision ensemble models for the prediction of the crude oil price and volatility, which can be broadly categorized into two types. One type combines data preprocessing methods with forecasting models. Before conducting predictive experiments, time series data is transformed into multiple modal functions using wavelet decomposition, variational mode decomposition (VMD), empirical mode decomposition (EMD) and their extended methods (Zhu et al., 2017; Wang et al., 2018; Liu et al., 2020b; Wang et al., 2021; Luo et al., 2021), fully extracting the mechanistic features of the data to improve predictive accuracy. At the same time, the other type is to combine various individual prediction models into a new predictive model, which can be more efficient (Lu et al., 2021; Ozdemir et al., 2022; Liu et al., 2023). Particularly, the attention mechanism, known for its ability to compute attention probability distributions to extract crucial information and optimize predictive models, has been widely adopted. In addition, Ouyang et al. (2021) confirmed that attention mechanism significantly improves the accuracy of time series forecasting.

The existing decomposition methods mainly include wavelet transform, VMD, EMD, as well as extended methods such as ensemble empirical mode decomposition (EEMD) and complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN). Their basic idea is to convert the original series into the sum of several subsequences, but the decomposition effects vary. Table 1 presents recent studies on crude oil price decomposition. Lin et al. (2022) indicated that wavelet transform decomposition can decompose crude oil price into low-frequency main signals and several high-frequency noise signals. Fang et al. (2023) found that EMD can decompose crude oil price into multiple regular intrinsic modes, making crude oil price prediction simpler and more accurate, but the decomposition process may suffer from endpoint effects, and the original variables might be overlooked. Therefore, some scholars have applied EEMD, CEEMDAN and VMD to the prediction of crude oil price, achieving significant improvements. EEMD can reduce the impact of noise during the decomposition process (Wu et al., 2019; Liu et al., 2020a; Jiang et al., 2022), CEEMDAN can decompose crude oil price into several intrinsic mode functions (Li et al., 2022), and VMD can suppress mode aliasing by controlling the bandwidth (Li et al., 2021b; Guo et al., 2022). However, these methods face issues such as high complexity and unreasonable parameter settings, which can affect prediction results (Xu and Niu, 2023).

According to the review of existing literature, we found that the improvement of crude oil price and volatility forecasting methods can be approached from two aspects: data preprocessing and deep learning.

In terms of data preprocessing improvement, many studies utilize methods such as EMD, EEMD and CEEMDAN, which have certain limitations. The extreme-point symmetric empirical mode decomposition (ESMD), overcoming the mode mixing issue of EMD and the high complexity problem of CEEMDAN, has been less applied in the research of crude oil market prediction. Regarding the improvement in deep learning, most ensemble models focus on the synthesis of traditional economics models with machine learning or neural networks. Few researchers consider combining neural networks with attention mechanisms for trends prediction in the crude oil market. Therefore, this study integrates ESMD, K-means clustering, fast independent component analysis (FastICA), BiLSTM, and attention mechanism, proposing a novel integrated model for the point prediction and interval prediction of crude oil price and volatility. To be more specific, Firstly, ESMD is utilized to decompose crude oil price and volatility into multiple modal functions, effectively extracting information from the data. Secondly, K-means clustering and FastICA are employed for data reconstruction and extraction of independent components, thereby reducing the correlation between sequences. Lastly, using the independent components as input, a BiLSTM-Attention composite model is constructed for the point prediction and interval prediction of crude oil price and volatility.

The main contributions of this study are described as follows: (1) We employ the ESMD method for the prediction of crude oil price and volatility, improving the decomposition performance of data asymmetry around the midpoint every half cycle and avoiding mode mixing issues present in methods like EMD. (2) K-means clustering and FastICA are utilized to reconstruct the decomposed data, enhancing prediction efficiency by fully exploring data features. (3) Building upon BiLSTM predictions, an attention mechanism is incorporated to assign weights, recognizing important information and enhancing the accuracy of both point and interval predictions.

The remainder of this study is conducted as follows. Section 2 presents the methodology used in this study, Section 3 covers data decomposition and reconstruction, Section 4 involves forecasting results, and Section 5 provides the conclusion.

2. Methodology

2.1. Research framework

The new integrated model we propose consists of five fundamental methods, namely extreme-point symmetry empirical mode decomposition (ESMD), K-means clustering, fast independent component analysis (FastICA), bidirectional long short-term memory network (BiLSTM) and attention mechanism. First, ESMD first proposed by Wang and Li (2013), is an adaptive data analysis method and can analyze a non-stationary and nonlinear signal, which is based on the natural phenomenon that a high-frequency small wave rides on a low-frequency bug wave and the small one is almost symmetric about its crest. We use ESMD to decompose crude oil price and volatility into multiple intrinsic modes. Second, K-means clustering automatically assigns similar samples to the same category. Considering the quality of ESMD decomposition results and prediction efficiency, multiple intrinsic modes are reconstructed through this method, grouping similar data together to optimize the decomposition results. Third, fast independent component analysis (FastICA) extracts independent components by searching for directions characterized by maximal non-Gaussianity. Independent component analysis assumes that the components are mutually independent, aiming to extract the intrinsic structure and features of data by separating independents from multidimensional signals. We further reconstruct the clustered data with FastICA, effectively mining sequence information while reducing data noise. Fourth, the long short-term memory neural network (LSTM) introduces gating mechanisms to delete or add past long-term information, while BiLSTM integrates forward and backward LSTM to extract features from both past and future information in time series. Therefore, we apply them to predict

Table 1
Recent studies on oil price decomposition.

Author(s)	Decomposition model	Main finding
Lin et al. (2022)	Wavelet transform	Wavelet transform decomposition can decompose crude oil price into low-frequency main signal and several high-frequency noise signals.
Fang et al. (2023)	EMD	EMD can decompose crude oil price into multiple regular intrinsic modes, making crude oil price forecasting simpler and more accurate. However, the decomposition process may encounter endpoint anomalies, and the original variables might be overlooked.
Wu et al. (2019) Liu et al. (2020a) Jiang et al. (2022)	EEMD	EEMD can reduce the impact of noise during the decomposition process. When combined with deep learning models such as LSTM and GRU, it can provide nearly perfect predictions of the crude oil market and demonstrate significant advantages.
Li et al. (2022)	CEEMDAN	CEEMDAN can decompose crude oil price into several intrinsic mode functions, but the resulting high complexity can negatively impact the prediction outcomes.
Li et al. (2021b) Guo et al. (2022) Xu and Niu (2023)	VMD	Models using VMD demonstrate significantly better prediction performance compared to those without VMD. Unlike the principle of EMD, VMD can suppress the mode aliasing phenomenon by controlling the bandwidth. However, determining the number of components for VMD remains challenging.

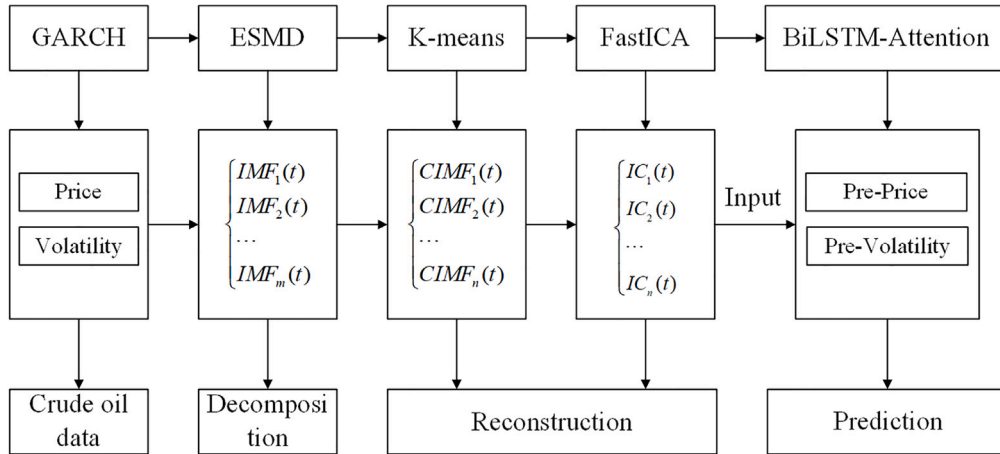


Fig. 1. Framework of the research.

crude oil price and volatility. Fifth, attention mechanism mimics the human attention principle to extract key information. We combine the attention with BiLSTM to assign weights to the mined information and enhance the prediction accuracy of crude oil price and volatility. Fig. 1 illustrates the specific prediction process. We define the crude oil price and volatility as the predicted variables, using ESMD to decompose them into intrinsic modes. Subsequently, we further reconstruct the decomposed data and extract independent components using the combination of K-means clustering and FastICA (we referred to as CFastICA in the following text). Then, we utilize these independent components as input through the BiLSTM-Attention mode to predict crude oil price and volatility. Sections 2.2–2.6 will introduce these specific methods in detail.

2.2. Extreme-point symmetric empirical mode decomposition (ESMD)

ESMD extends the definition of intrinsic modes without the need for denoising, noise addition, smoothing, or other processing methods, achieving direct data decomposition, addressing the mode mixing issue in traditional empirical mode decomposition by interpolating the midpoints of line segments connecting local maxima and minima of the data. The specific computational steps are as follows:

Step 1: Identify all local extremal points E_i ($1 \leq i \leq n$) of the data X , including both maxima and minima. Connect all adjacent points E_i and label their midpoints F_i ($1 \leq i \leq n-1$) with line segments. Simultaneously, supplement a left boundary midpoint F_0 and a right boundary midpoint F_n using linear interpolation.

Step 2: Construct p interpolation curves $L_1, L_2, \dots, L_p$ ($p \geq 1$) using $n+1$ midpoints F_i ($i = 1, 2, \dots, n$), at the same time, calculate the average curve L^* , where $L^* = \frac{L_1 + L_2 + \dots + L_p}{p}$.

Step 3: Repeatedly perform steps one and two for $X - L^*$, and if the condition $|L^*| \leq \varepsilon$ is satisfied (where ε is the pre-set tolerance and σ_0 is the standard deviation of data X , typically chosen as $\varepsilon = 0.0001\sigma_0$), or the iteration reaches the pre-set maximum value of K , obtain the first mode $IMF_1 = m_1$.

Step 4: Repeat steps 1 to 3 for $X - m_1$, obtaining subsequent modes $m_2, m_3, \dots$, until the number of extremal points contained in the residual function $R = X - m_1 - m_2 - \dots - m_c$ reaches a pre-set value.

Step 5: Change the value of K sequentially according to a pre-defined integer interval $[K_{min}, K_{max}]$, and repeat the above steps. Determine $K = K_0$ based on the minimum variance ratio σ/σ_0 (σ and σ_0 represent respectively the standard deviations of $X - R$ and X). Repeat steps 1 to 4 to obtain the ultimate decomposition result.

2.3. K-means clustering

The K-means algorithm, one of the most widely used unsupervised learning methods, effectively partitions a dataset into k clusters. It ensures that each data point is assigned to the nearest cluster, with the cluster center being the average of all data points within it. This algorithm relies on iterative optimization, wherein each iteration updates the cluster centroids until a convergence condition is satisfied (Chen and Wen, 2008; Sinaga and Yang, 2020). Its clustering steps are as follows:

- (1) Initialization. Choose to partition the dataset into k clusters, then randomly select k data points as initial cluster centers.
- (2) Assignment. Assign each data point to the cluster center closest to it, and each data point can only belong to one cluster.
- (3) Update. Update the cluster centers based on the assigned data points, which is achieved by calculating the average of the data points belonging to each cluster.
- (4) Repeat. Repeat steps 2 and 3 until the cluster centers no longer change, or the predefined number of iterations is reached.
- (5) Output. Obtain k clusters and the center of each cluster.

2.4. Fast independent component analysis (FastICA)

Independent component analysis (ICA) posits that n random observed signals $X = (x_1, x_2, \dots, x_n)$ can be linearly composed of p ($p \leq n$) random and mutually independent signals $S = (s_1, s_2, \dots, s_p)$, satisfying the equation:

$$x_i = a_{i1}s_1 + a_{i2}s_2 + \dots + a_{ip}s_p, i = 1, 2, \dots, n, \quad (1)$$

where $a_{i1}, a_{i2}, \dots, a_{ip}$ are the coefficients. Then, the expression in Eq. (1) can be transformed into matrix form as:

$$X = AS, \quad (2)$$

where X is the known observed signal matrix, S indicates the unknown independent source vector, and A is the unknown mixing matrix. To improve the accuracy of independent vector separation, a separation matrix, W is constructed. If Y is the optimal signal estimate, it satisfies:

$$Y = WX = WAS. \quad (3)$$

Therefore, the key to ICA is the solution for the separation matrix W . Through the examination by Hyvärinen and Oja (2000), the fast independent component analysis algorithm (FastICA) converges quickly and does not require setting a step size parameter. So, we choose this method and, based on the criterion of maximizing non-Gaussianity with respect to $W^T X$, solves the separation matrix W . Specifically, selecting an appropriate nonlinear function g , which satisfies:

$$g(y) = \tanh(by), \quad (4)$$

here, y is a subvector in the signal Y , $b = 1$. Simultaneously, formulate the optimization problem of maximizing negentropy as:

$$f_g(W^T X) = (E[g(W^T X)] - E[g(m)])^2, \quad (5)$$

where $f_g(W^T X)$ represents negentropy, $E[\cdot]$ denotes the mean function, m is the centralized and whitened Gaussian random variable. To solve Eq. (5), according to the Kuhn–Tucker conditions, given the constraints:

$$E[(W^T X)^2] = \|W\|^2 = 1, \quad (6)$$

and the optimal value calculation can be expressed as:

$$\begin{cases} E[Xg'(W^T X)] - \gamma W = 0 \\ \gamma = E[W_0^T Xg'(W_0^T X)] \end{cases}, \quad (7)$$

where $g'(\cdot)$ is the derivative of $g(\cdot)$, W_0 is the optimal value of W , and γ is a constant.

2.5. Bidirectional long short-term memory network (BiLSTM)

The cell unit of the LSTM consists of a memory storage (Cell) and three gated settings (Gates), which is used to store the neuron state. The input gate and output gate are responsible for receiving, outputting parameters, and adjusting parameters. The forget gate controls the degree to which the previous cell state is forgotten. The specific working principles are explained as follows. And the constructed BiLSTM in this paper consists of two independent LSTMs, namely the forward LSTM and the backward LSTM. Simultaneously considering information from past and future sequences, the final learning result is obtained by combining the forward input sequence and the reverse input sequence with certain weights. The specific model structure is illustrated in Fig. 2.

To be specific, if x_{t-1} , x_t and x_{t+1} are input variables, after going through the forward LSTM, we obtain hidden vectors $\tilde{h}_{t-1}$, $\tilde{h}_t$, $\tilde{h}_{t+1}$. At the same time, the same input sequences undergo the backward LSTM, yielding hidden vectors $\tilde{h}_{t-1}$, $\tilde{h}_t$ and $\tilde{h}_{t+1}$. Further, concatenating the forward hidden vectors with the backward hidden vectors results in the output of the BiLSTM, represented by h_{t-1} , h_t and h_{t+1} .

2.6. Attention mechanism

The attention mechanism, as a mechanism to enhance the effectiveness of recurrent neural networks, has been widely used in various fields of deep learning, particularly in sequence data such as speech recognition and machine translation. Similar to the attention resource allocation mechanism of the human brain, the attention mechanism assigns probabilistic weights to input elements, focusing more on key information and reducing the impact and interference of irrelevant information, thereby improving prediction accuracy. In this study, we integrate the attention mechanism with BiLSTM, assigning different weights to the hidden states at different time steps, which helps the model to weigh the importance of the input sequence. Fig. 3 illustrates the structure of the attention mechanism.

Here, $h_1, h_2, \dots, h_{t-1}, h_t$ represent the input vectors to the BiLSTM. We define q as the query vector, V as the value vector, W and b as the learned weight matrix and bias vector, respectively. By using $s(h_i, q) = V^T \tanh(W h_i + b)$, we look up and select certain information from the input data. Further, based on $\rho_i = \text{softmax}(s(h_i, q)) = \exp(s(h_i, q)) / \sum_{i=1}^t \exp(s(h_i, q))$, the attention scores are converted into attention weights, and finally, we obtain $y = \sum_{i=1}^t \rho_i h_i$.

3. Data decomposition and reconstruction

Crude oil price and volatility, as a type of time series, exhibit highly uncertain, nonlinear, and volatile characteristics. To fully exploit the information embedded in these data, this section employs ESMD-CFastICA for data decomposition and reconstruction. Using ESMD, the sequences are decomposed into multiple intrinsic modes. Subsequently, CFastICA is applied for cluster reconstruction and independent component analysis on the decomposed data, extracting multiple mutually independent signals from the original sequences for prediction purposes.

3.1. Data selection and description

WTI crude oil is one of the three major pricing benchmarks in the global oil market and is commonly used as a proxy indicator for international crude oil market dynamics (Zhang et al., 2023b). In this study, daily WTI crude oil price from April 4, 1983, to December 12, 2023, is selected, totaling 10,218 data points. On this basis, daily logarithmic return is calculated, followed by unit root tests and Arch effects tests. The results indicate that the return is significantly stable at the 1% level and exhibit arch effects. Furthermore, a GARCH(1,1)

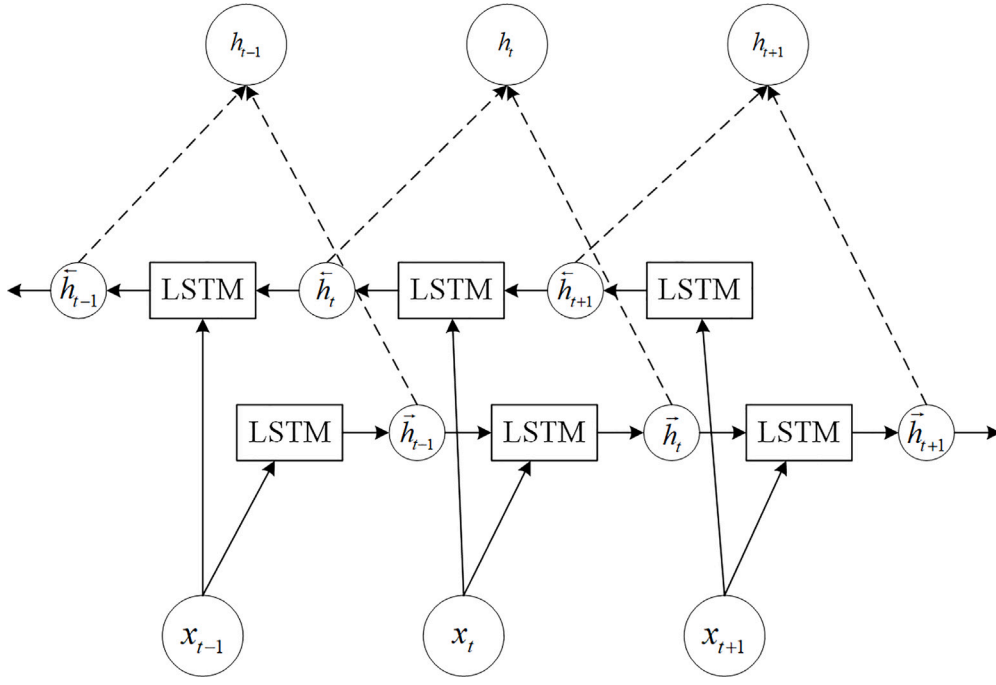


Fig. 2. The model structure of BiLSTM.

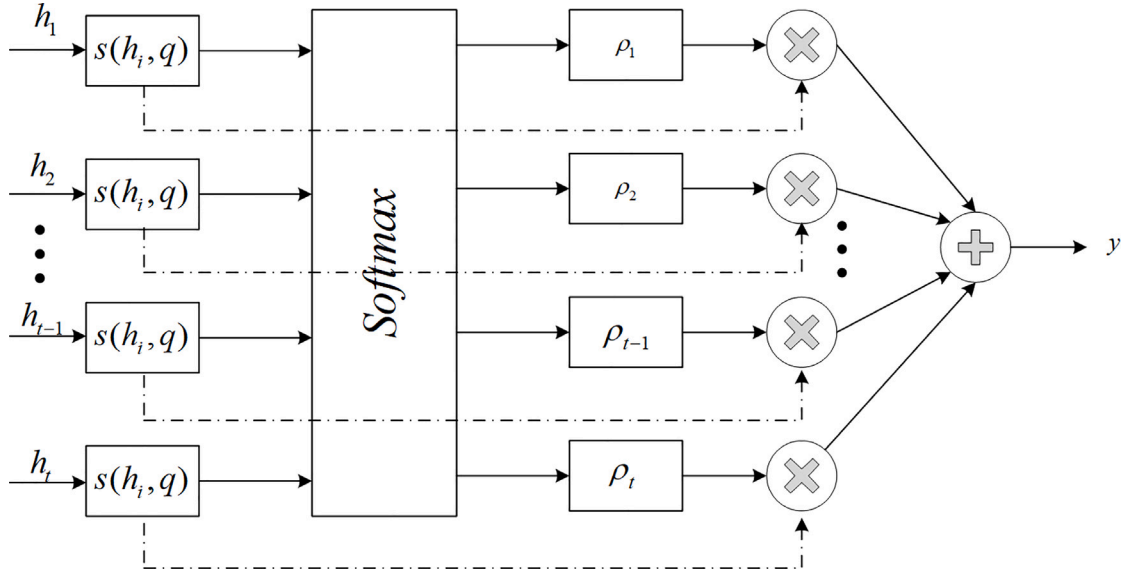


Fig. 3. The structure of attention mechanism.

model is constructed to calculate volatility, with the specific model as follows:

$$\sigma_t^2 = \omega + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad (8)$$

where σ_t^2 presents the volatility at time t , ω , α_1 and β_1 are the parameters to be estimated. And the results show that they are all significantly at the 1% level.

The time series plot of WTI crude oil price and volatility is shown in Fig. 4, where WTI-price and WTI-volatility represent WTI crude oil price and volatility, respectively. As shown in the figure, the overall trend of WTI crude oil price has been upward from 1983 to the present, accompanied by fluctuations. After 2000, the growth rate of WTI price increased significantly compared to before 2000. Specifically, before 2000, the volatility of WTI-price was evident during the energy crises caused by the 1990 Gulf War and the 2000 California electricity crisis,

indicating a highly unstable global oil market at that time. After 2000, with rapid global economic development, the international oil market experienced drastic fluctuations influenced by events such as the 2008 financial crisis and the 2022 Russia-Ukraine war, highlighting the impact of extreme events and geopolitical conflicts on the oil market. In terms of WTI-volatility, the most significant changes occurred in 2022, followed by 1990, indicating the substantial impact of the Russia-Ukraine war in 2022 on the oil market.

3.2. Decomposition and reconstruction

Before predicting WTI crude oil price and volatility, this study conducts preprocessing on data to fully exploit the information, namely, decomposition and reconstruction. To reconstruct the data, the first step involves using the relevant procedures of ESMD to decompose

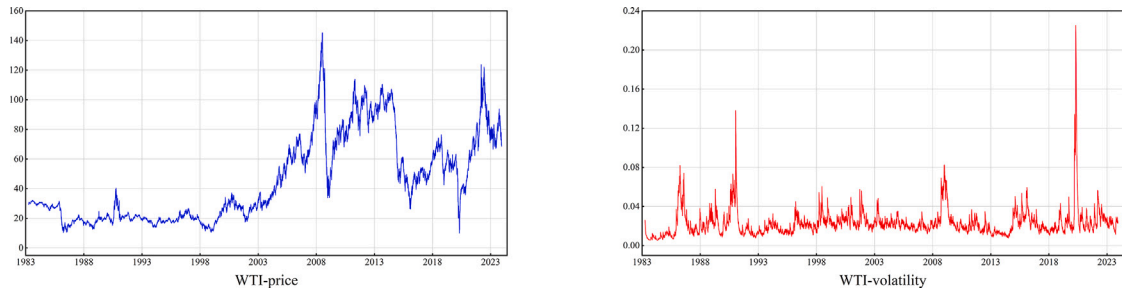


Fig. 4. The time series of WTI crude oil price and volatility.

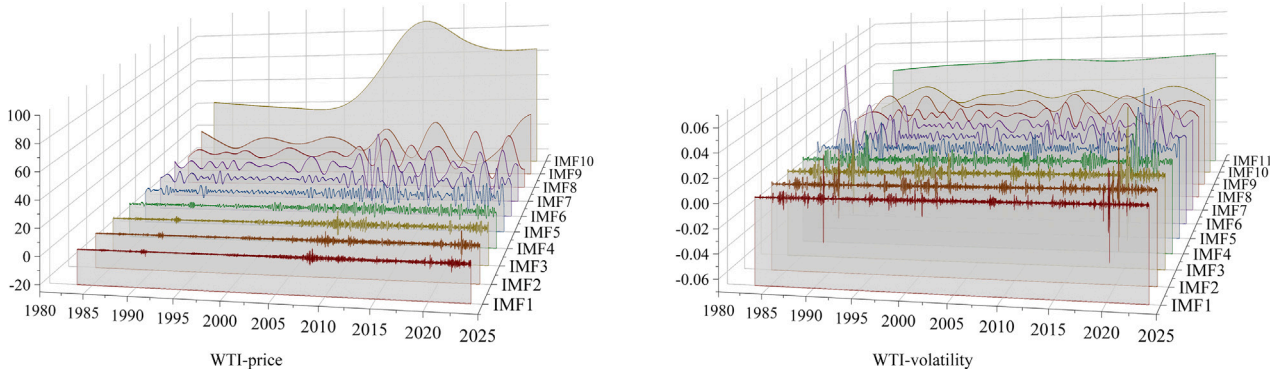


Fig. 5. The decomposition results for WTI crude oil price and volatility.

WTI-price and WTI-volatility into 10 and 11 intrinsic modes, respectively. Fig. 5 presents the decomposition results for WTI-price and WTI-volatility. And the horizontal axis represents time, the vertical axis indicates intrinsic mode numbers, and the values of the modal functions are presented on the vertical coordinate.

The decomposition results for WTI crude oil price and volatility in Fig. 5 can be classified into three categories based on the fluctuation frequencies of individual sub-series: high frequency, medium frequency, and low frequency. Specifically, IMF1 to IMF3 of WTI-price can be considered high-frequency data, IMF4 to IMF9 as medium-frequency data, and IMF10 as low-frequency data. For WTI-volatility, the delineation is made using IMF4 and IMF10 as boundaries. In detail, IMF1 to IMF3 exhibit the highest fluctuation frequencies, IMF4 to IMF9 are considered medium-frequency, and IMF10 and IMF11 represent low-frequency data, with their variations tending to be horizontal. Overall, the frequency of IMF decreases with the increase in the index, and the last intrinsic mode exhibits very smooth variations.

Although ESMO resolves the mode mixing problem introduced by EMD, its sensitivity to noise can lead to the instability of the decomposition results. To enhance the predictive efficiency of the data, in this paper, the reconstructed data is obtained using CFastICA, a combination of K-means clustering and the FastICA model. CFastICA is employed to ensure the utilization of the minimum number of sub-series while fully exploiting the information present in the data. Firstly, K-means clustering is applied separately to all intrinsic modes of WTI-price and WTI-volatility. The elbow method is utilized to determine the number of clusters. This study comprehensively assesses the clustering numbers for WTI-price and WTI-volatility using the average within-cluster sum of squares (WCSS), sum of squared errors (SSE), and the silhouette coefficient (SC).

Fig. 6 shows the change in the number of clusters for WTI-price and WTI-volatility corresponding to the values of WCSS, SSE, and SC. The horizontal axis represents the values of the clustering number k , and the vertical axis represents the values of WCSS, SSE, and SC. The red markers indicate the finally selected number of clusters. The criterion for selecting the number of clusters is based on the fastest rate of

Table 2

The category to each intrinsic mode belongs.

WTI	CIMF	IMF
Price	CIMF1	IMF1~IMF8
	CIMF2	IMF9
	CIMF3	IMF10
	CIMF1	IMF1~IMF7
Volatility	CIMF2	IMF8
	CIMF3	IMF9
	CIMF4	IMF10
	CIMF5	IMF11

decrease in WCSS and SSE, as well as the situation where the SC reaches its peak. By observation, the WCSS and SSE for WTI-price show the fastest decrease rates at $k = 2$ and 3, with SC showing less distinct performance. Therefore, the intermediate value $k = 3$ is chosen for WTI-price. Similarly, for WTI-volatility, the best values for WCSS, SSE, and SC are observed at $k = 5$. Hence, the final determined number of clusters for WTI-price and WTI-volatility are 3 and 5, respectively.

Table 2 shows the categories to which each intrinsic mode belongs. Summing up the modes within the same category, the reconstructed data for WTI-price and WTI-volatility are denoted as CIMF1 ~ CIMF3 and CIMF1 ~ CIMF5, respectively. Subsequently, the FastICA method is employed to extract independent signals from the CIMF sub-series by solving the optimization problem that maximizes negative entropy. The estimated mixing matrix is obtained, leading to the extraction of independent signals. Through multiple decomposition experiments, this study selects the number of independent components (IC) as 3.

Fig. 7 displays the temporal plots of independent components for WTI-price and WTI-volatility, respectively. We observe that IC1 of WTI-price and WTI-volatility exhibits a pattern highly similar to the original sequences. IC1 can be considered as a reflection of the inherent characteristics of WTI-price and WTI-volatility, indicating the significant influence of historical data on their variations. In contrast, the variations in IC2 are smoother but accurately identify the impact of major international events. Therefore, IC2 is regarded as representing

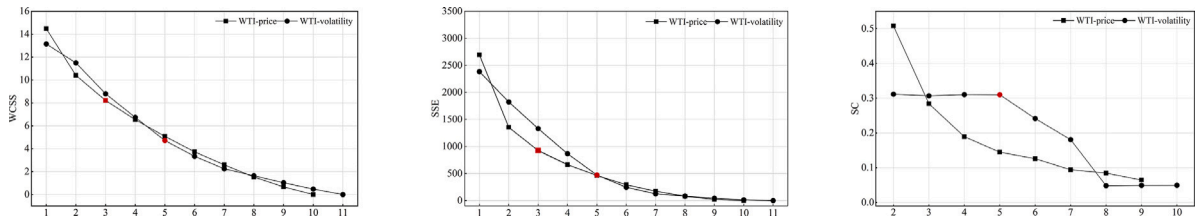


Fig. 6. Different numbers of clusters lead to varying values of WCSS, SSE, and SC.

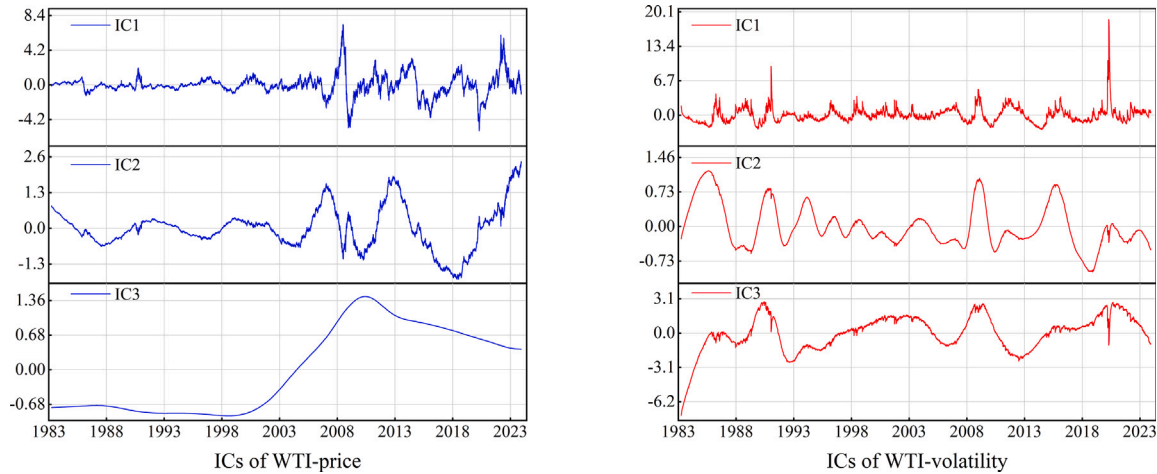


Fig. 7. Independent components extraction via CFastICA.

Table 3

Non- Gaussian distribution test for independent components.

WTI	IC	J-B test	A-D test	K-S test
Price	IC1	11081.0000***	315.7000***	0.0866***
	IC2	1185.5000***	690.1100***	0.2073***
	IC3	463.5200***	116.6000***	0.1469***
Volatility	IC1	328729.0000***	164.5500***	0.0821***
	IC2	950.4000***	269.4300***	0.2471***
	IC3	3381.6000***	54.4560***	0.1329***

Note: *, ** and *** represent significance at the 0.1, 0.05, and 0.01 levels, respectively.

the influence of global shocks on WTI-price and WTI-volatility. The variations in IC3 are considered as reflecting changes in WTI-price and WTI-volatility induced by other factors.

Independent component analysis (ICA) requires that the extracted signals exhibit non-Gaussian distribution and independence. Non-Gaussian distribution is assessed first in this study using Jarque–Bera (J-B test), Anderson–Darling (A-D test), and Kolmogorov–Smirnov (K-S test) to examine whether each independent component follows a normal distribution. The results in Table 3 indicate that all independent components (ICs) do not follow a normal distribution, and they are statistically significant at the 1% level, confirming the non-Gaussian nature of the ICs.

Then, a correlation test is conducted. Fig. 8 shows the heatmap of the correlation coefficients of the independent components of WTI-price and WTI-volatility. The colors range from blue to red, representing a gradual transition from −1 to 1. Deeper red indicates a stronger positive correlation, while deeper blue indicates a stronger negative correlation. The diagonal line appears in deep red, indicating a high positive correlation between each independent component and itself. The low correlation between different independent components suggests that the reconstructed data has independence.

4. Forecasting results

After decomposing and reconstructing the data, this section combines BiLSTM with the attention mechanism to forecast crude oil price and volatility. On one hand, the model performance evaluation criteria are established from the perspectives of point prediction and interval prediction, including model accuracy evaluation, model effectiveness evaluation, and model significance evaluation. On the other hand, by comparing different preprocessing methods, machine learning and deep learning models, and their combination models, the optimal performance of the proposed novel integrated model, ESMD-CFastICA-BiLSTM-Attention model, in the prediction of crude oil price and volatility is verified.

4.1. Performance criteria

In recent years, a series of effective indicators have been used for the performance evaluation of prediction models. We construct an evaluation system from three perspectives: model accuracy evaluation, model effectiveness evaluation, and model significance evaluation. Specifically, model accuracy evaluation is categorized into point prediction and interval prediction. For point prediction, four metrics are selected: mean absolute error (MAE), root mean square error (RMSE), goodness of fit (R^2), and directional prediction statistic (D_{stat}). Among them, MAE reflects the actual situation of prediction errors, RMSE effectively reflects the differences between actual values and predicted values, R^2 focuses on revealing the dispersion differences between predicted values and true values, and D_{stat} primarily reflects the accuracy of the predicted direction. In additionally, we select three metrics, namely, prediction interval coverage probability (PICP), prediction interval average width (PIAW) and accumulated width deviation (AWD), to reflect the performance of interval prediction. The PICP indicator measures the probability that the true observed values fall within the predicted interval, reflecting the reliability of the interval prediction results. A higher PICP value indicates higher accuracy in interval prediction.

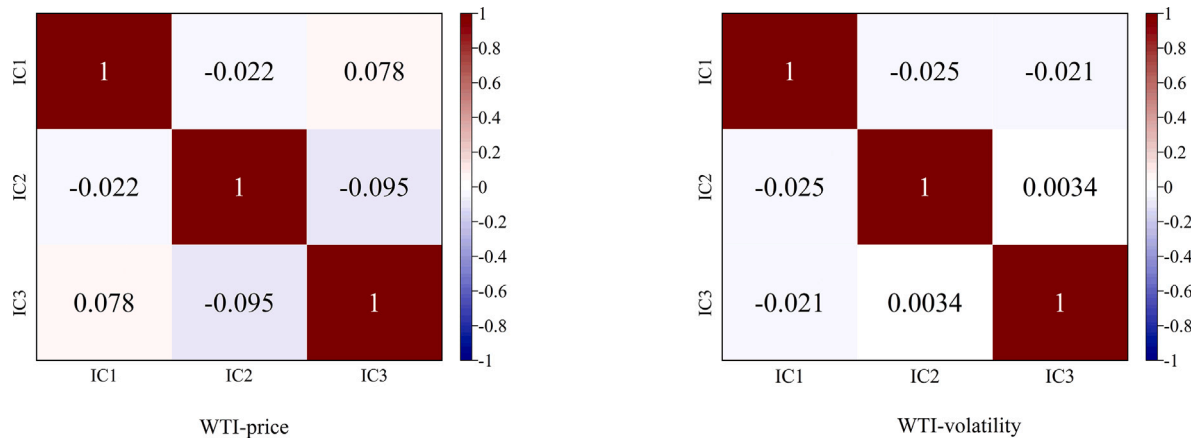


Fig. 8. Independent components correlation coefficient.

Table 4

The calculation formulas for metrics of point and interval prediction.

Prediction	Metrics	Calculation formulas
Point	MAE	$MAE = \frac{1}{N} \sum_{i=1}^N \hat{x}_i - x_i $
	RMSE	$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2}$
	R^2	$R^2 = 1 - \frac{\sum_{i=1}^N (x_i - \bar{x}_i)^2}{\sum_{i=1}^N (x_i - \hat{x}_i)^2}$
	D_{stat}	$D_{stat} = \frac{1}{N} \sum_{i=1}^N r(i) \times 100\%, r(i) = \begin{cases} 1, (x_{i+1} - x_i)(\hat{x}_{i+1} - \hat{x}_i) \geq 0 \\ 0, (x_{i+1} - x_i)(\hat{x}_{i+1} - \hat{x}_i) \leq 0 \end{cases}$
Interval	PICP	$PICP = \frac{1}{N} \sum_{i=1}^N c_i$
	PIAW	$PIAW = \frac{1}{N} \sum_{i=1}^N (U_i - L_i)$
	AWD	$AWD = \frac{1}{N} \sum_{i=1}^N awd_i, awd_i = \begin{cases} \frac{L_i - \hat{x}_i}{U_i - L_i}, x_i < L_i \\ 0, x_i \in [L_i, U_i] \\ \frac{\hat{x}_i - U_i}{U_i - L_i}, x_i > U_i \end{cases}$

Note: N represents the data quantity, x_i denotes the true values, $\hat{x}_i$ stands for the predicted values, and $\bar{x}_i$ represents the mean value. U_i and L_i denote the upper and lower limits of the prediction interval. If $x_i \in [L_i, U_i]$, then $c_i = 1$, otherwise, its value is 0.

Table 5

The calculation of model effectiveness evaluation metrics.

Indicators	Meaning	Formulas
I_{MAE}	The improvement percentage of MAE	$I_{MAE} = \left \frac{MAE_1 - MAE_2}{MAE_1} \right \times 100\%$
I_{RMSE}	The improvement percentage of RMSE	$I_{RMSE} = \left \frac{RMSE_1 - RMSE_2}{RMSE_1} \right \times 100\%$

The PIAW effectively reflects the width of the prediction interval, with smaller values indicating better predictive performance. The AWD indicator does not consider the deviation of the actual observed values falling within the predicted interval, it only measures the deviation of observed values outside the predicted interval. A smaller AWD value indicates better interval prediction performance. Table 4 provides the calculation formulas for these indicators.

While MAE and RMSE can effectively measure the performance of prediction models, in order to compare the predictive effectiveness between different models, We employ the improvement percentage of MAE and RMSE between two models for model evaluation. The specific calculation formulas are presented in Table 5. MAE_1 represents the MAE value of Model 1, MAE_2 corresponds to the MAE value of Model 2; $RMSE_1$ denotes the RMSE value of Model 1, and $RMSE_2$ stands for the RMSE value of Model 2.

To assess the significant differences between predictive models, besides measuring model accuracy, it is essential to determine whether there are significant differences in the predictive results of two models at a given significance level. Traditional methods like F-test and MR statistic have certain limitations and may not capture the significance differences between models effectively. Therefore, this study

employs the Diebold–Mariano test (DM test) to evaluate the significance of model prediction results. This test method can compare whether the predictive results of different models have statistically significant differences. The prediction errors of two models are:

$$e_{t+l}^1 = x_{t+l} - \hat{x}_{t+l}^1, l = 1, 2, \dots, m, \quad (9)$$

$$e_{t+l}^2 = x_{t+l} - \hat{x}_{t+l}^2, l = 1, 2, \dots, m. \quad (10)$$

Defining the loss function as $F(e_{t+l}^i), i = 1, 2$, used to evaluate the predictive accuracy of the model. Further, constructing the MD test statistic as:

$$DM = \frac{\sum_{l=1}^m (F(e_{t+l}^1) - F(e_{t+l}^2)) / m}{\sqrt{S^2 / m}}, \quad (11)$$

where S^2 is the variance estimation of $d_l = F(e_{t+l}^1) - F(e_{t+l}^2)$. Given the null hypothesis $H_0 : E(d_l) = 0$, which states that the two models have the same predictive performance, and the alternative hypothesis $H_1 : E(d_l) \neq 0$, which suggests that the two models have different predictive performance. If $|DM| > z_{\alpha/2}$ (where $z_{\alpha/2}$ represents the z -value of the standard normal distribution at the significance level α),

Table 6
The composition of all integrated models.

Category	Model	ES-CF-BiAt	ES-CF-BiLSTM	ES-CF-LSTM	ES-CF-SVR	ES-CF-RF	EE-CF-BiAt	E-CF-BiAt	ES-BiAt	S-BiAt
Decomposition	EMD							YES		
	EEMD						YES			
	ESMD	YES	YES	YES	YES	YES			YES	
Reconstruction	Clustering	YES	YES	YES	YES	YES	YES	YES		
	FastICA	YES	YES	YES	YES	YES	YES	YES		
	BiLSTM	YES	YES				YES	YES	YES	YES
Deep Learning	LSTM			YES						
	SVR				YES					
	RF					YES				
	Attention	YES					YES	YES	YES	YES

the null hypothesis is rejected, indicating a significant difference in predictive results between the two models. Otherwise, the null hypothesis cannot be rejected, suggesting no significant difference between the two models.

4.2. Prediction outcomes

In this section, we investigate the superiority of the new integrated model ESMD-CFastICA-BiLSTM-Attention in predicting crude oil price and volatility from two perspectives: point prediction and interval prediction. Firstly, we use the ESMD model to decompose oil price and volatility into multiple intrinsic modes. Next, we perform reconstruction and independent component extraction through k-means clustering and FastICA. Subsequently, using the independent components as input, we employ a combination model of BiLSTM and Attention for point prediction and result evaluation. Finally, we combine the point prediction results with the distribution characteristics of crude oil price and volatility for interval prediction and result assessment. Meanwhile, we introduce two decomposition models, empirical mode decomposition (EMD) and ensemble empirical mode decomposition (EEMD), as well as four deep learning models including bidirectional long short-term memory neural network (BiLSTM), long short-term memory neural network (LSTM), support vector regression (SVR), and random forest (RF). We construct various combination models to predict crude oil price and volatility and compare the prediction results with our proposed combination model.

For ease of reference in subsequent sections, Table 6 summarizes all the basic models we used, categorized into three groups: decomposition, reconstruction, and deep learning methods, and assigns specific names to the combination models. In the table, “YES” indicates that the basic model is included in the combination model, while blank spaces indicate that the basic model is not included. ES-CF-BiAt represents the approach of decomposing the predictive variables using ESMD, followed by reconstruction and independent component extraction using k-means clustering and FastICA, and finally prediction using BiLSTM and Attention. In comparison to ES-CF-BiAt, ES-CF-BiLSTM, ES-CF-LSTM, ES-CF-SVR, and ES-CF-RF involve only variations in the deep learning models, while EE-CF-BiAt and E-CF-BiAt involve only replacement of the decomposition methods. ES-BiAt does not involve reconstruction, and S-BiAt denotes direct prediction on the original sequence.

The time interval of experimental data for crude oil price and volatility is defined from April 5, 1983, to December 12, 2023. The dataset is split into training and testing sets with a ratio of 8:2. The training set spans from April 5, 1984, to October 29, 2015, while the testing set covers the period from October 30, 2015, to December 12, 2023. The WTI crude oil market operates five days a week, and the prediction is made for the next day's data based on the information available from the preceding five days. To eliminate the influence of data units, the data is normalized before prediction. Referencing the studies by Li et al. (2021a) and Ouyang et al. (2021), we initially configure the BiLSTM-Attention model with an input layer, a BiLSTM

layer, an attention layer, and an output layer. We then use the grid search method to determine the optimal parameter combination, with each layer having 20 nodes, the activation function set to “tanh”, the number of epochs and batch size set to 100, and the optimizer used is adam. Additionally, to directly compare the performance of BiLSTM, LSTM, and BiLSTM-Attention in predicting crude oil prices and volatility, we set the hyperparameters of BiLSTM and LSTM to be the same as those of BiLSTM-Attention. Furthermore, we configure the SVR model with the rbf kernel function and the RF model with 100 n_estimators. Other parameter settings are detailed in Table 7.

After determining the optimal parameters, we conducted point prediction for crude oil price and volatility using ES-CF-BiAt. Fig. 9 presents the comparison between the real values and the predicted values. The blue dashed line and red dashed line represent the predicted values of crude oil price and volatility obtained using ES-CF-BiAt, respectively, while the black solid line represents the real values. We observed that, except for some extreme values, the majority of predicted values exhibit very close trends and magnitudes to the actual values for both datasets. Therefore, we preliminarily conclude that the selected model parameters are reasonable and can effectively predict crude oil price and volatility.

Table 8 presents the point prediction results for crude oil price and volatility using ES-CF-BiAt, EE-CF-BiAt, E-CF-BiAt, ES-BiAt, and S-BiAt. Bold font indicates the optimal results, and two sets of comparative experiments are highlighted to emphasize the importance of ESMD decomposition and CFastICA reconstruction in forecasting crude oil price and volatility. Firstly, the results of ES-CF-BiAt are compared with those of EE-CF-BiAt and E-CF-BiAt, where the difference lies in the decomposition method used. The point prediction results for crude oil price using ES-CF-BiAt are MAE = 1.2779, RMSE = 2.0835, $R^2 = 0.9877$, and $D_{stat} = 0.7204$. Both MAE and RMSE are lower than those of the EE-CF-BiAt and E-CF-BiAt models, while R^2 and D_{stat} are higher than those of the two comparative models. Similarly, the comparison results for crude oil volatility also follow this pattern except for D_{stat} . Thus, we find that ES-CF-BiAt yields the best point prediction results for crude oil price and volatility, indicating that the performance of ESMD decomposition is superior to that of EMD and EEMD. Secondly, ES-CF-BiAt is compared with S-BiAt, as the ES-CF-BiAt model decomposes and reconstructs data for crude oil price and volatility prediction, while S-BiAt directly predicts the original sequences. The comparison reveals that the prediction results for crude oil price and volatility using S-BiAt exhibit higher MAE and RMSE compared to ES-CF-BiAt, while R^2 and D_{stat} are lower than those of ES-CF-BiAt. This demonstrates that decomposing and reconstructing data through ESMD and CFastICA can enhance the prediction accuracy of crude oil price and volatility. In summary, we find that ESMD has better data decomposition capabilities compared to EMD and EEMD, and when combined with FastICA, it significantly improves the predictive performance of crude oil price and volatility.

To further examine the impact of BiLSTM-Attention on the prediction of crude oil price and volatility, we replaced BiLSTM-Attention with four other models: BiLSTM, LSTM, SVR, and RF, based on the decomposed and reconstructed data. Table 9 presents the four evaluation

Table 7
The hyperparameter tuning with different models.

Model	Hyperparameter tuning
BiLSTM-Attention	nodes = 20; activation = ‘‘tanh’’; epochs = 100; batch_size = 100; optimizer = ‘‘adam’’
BiLSTM	nodes = 20; activation = ‘‘tanh’’; epochs = 100; batch_size = 100; optimizer = ‘‘adam’’
LSTM	nodes = 20; activation = ‘‘tanh’’; epochs = 100; batch_size = 100; optimizer = ‘‘adam’’
SVR	kernel = ‘‘rbf’’; gamma = ‘‘auto’’
RF	n_estimators = 100; max_features = ‘‘auto’’; max_depth = 3; min_samples_split = 2; min_samples_leaf = 1

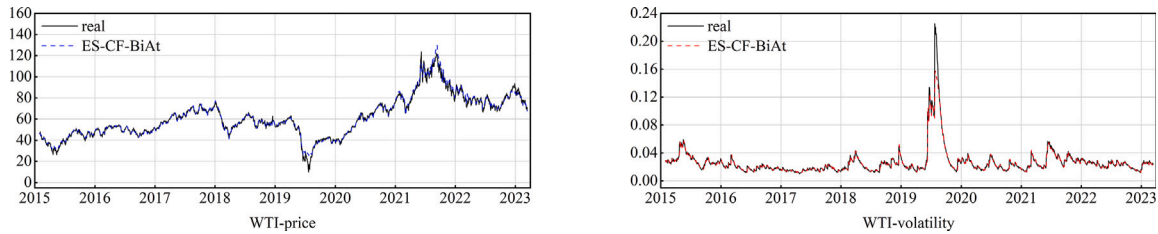


Fig. 9. The predicted values of crude oil price and volatility compared to the real values.

Table 8
Crude oil price and volatility prediction results with different data preprocessing models.

	Model	MAE	RMSE	R^2	D_{stat}
Price	ES-CF-BiAt	1.2779	2.0835	0.9877	0.7204
	EE-CF-BiAt	2.2395	2.8322	0.9773	0.4912
	E-CF-BiAt	1.5946	2.3555	0.9843	0.6190
	ES-BiAt	3.3284	4.0660	0.9533	0.6714
	S-BiAt	1.5961	2.2132	0.9862	0.4971
Volatility	ES-CF-BiAt	0.0017	0.0059	0.9252	0.5578
	EE-CF-BiAt	0.0029	0.0107	0.7523	0.5333
	E-CF-BiAt	0.0024	0.0090	0.8258	0.5646
	ES-BiAt	0.0027	0.0087	0.8347	0.6670
	S-BiAt	0.0028	0.0142	0.5590	0.5607

Table 9
Crude oil price and volatility prediction results with different machine learning models.

	Model	MAE	RMSE	R^2	D_{stat}
Price	ES-CF-BiAt	1.2779	2.0835	0.9877	0.7204
	ES-CF-BiLSTM	1.3354	1.9654	0.9891	0.4941
	ES-CF-LSTM	1.8335	2.5275	0.9820	0.4873
	ES-CF-SVR	13.9682	16.5110	0.2299	0.5093
	ES-CF-RF	8.7954	10.8565	0.6671	0.3972
	ES-CF-BiAt	0.0017	0.0059	0.9252	0.5578
	ES-CF-BiLSTM	0.0031	0.0142	0.5592	0.5622
Volatility	ES-CF-LSTM	0.0119	0.0243	-0.2792	0.5073
	ES-CF-SVR	0.0088	0.0122	0.6740	0.5602
	ES-CF-RF	0.0037	0.0122	0.6743	0.5083

metrics for crude oil price and volatility for each model. Among them, ES-CF-BiAt and ES-CF-BiLSTM perform better than the other models in predicting crude oil price and volatility. For crude oil price prediction, the two evaluation metrics RMSE and R^2 do not explicitly demonstrate the advantage of the Attention mechanism in ES-CF-BiAt. Similarly, in volatility prediction, the results do not sufficiently highlight the role of the attention mechanism. Therefore, further research is needed to evaluate the effectiveness of BiLSTM-Attention in predicting crude oil price and volatility.

Table 10 calculates the improvement percentage of ES-CF-BiAt relative to other comparative models in terms of MAE and RMSE, further evaluating the predictive effectiveness of different models. In the prediction of crude oil price, compared to ES-CF-SVR and ES-CF-RF, ES-CF-BiAt exhibits the largest improvement percentage, with both improvement metrics exceeding 400%. Additionally, we observe that the prediction result of ES-CF-BiAt are relatively close to those of ES-CF-BiLSTM for crude oil price prediction. However, ES-CF-BiAt shows a 4.50% improvement in MAE and a 5.67% improvement in RMSE relative to ES-CF-BiLSTM. Compared to crude oil price prediction,

Table 10
The improvement percentage of crude oil price and volatility.

Model	Price		Volatility	
	I_{MAE} (%)	I_{RMSE} (%)	I_{MAE} (%)	I_{RMSE} (%)
EE-CF-BiAt	75.25	35.94	76.70	82.04
E-CF-BiAt	24.78	13.06	46.36	52.68
ES-BiAt	160.45	95.15	61.93	48.71
S-BiAt	24.89	6.22	66.69	142.90
ES-CF-BiLSTM	4.50	5.67	86.04	142.85
ES-CF-LSTM	43.48	21.31	617.30	313.68
ES-CF-SVR	993.03	692.47	431.79	108.83
ES-CF-RF	588.25	421.07	125.20	108.74

the improvement percentage of using ES-CF-BiAt to predict crude oil volatility is more pronounced relative to other models, ranging from a minimum of $I_{MAE} = 46.36\%$ to $I_{RMSE} = 52.68\%$. Therefore, our proposed model demonstrates greater effectiveness in crude oil price and volatility prediction compared to other comparative models, resulting in a significant enhancement in predictive accuracy.

Furthermore, we employ the DM test to investigate whether there are statistically significant differences between ESMD-CFastICA-BiLSTM-Attention and other models at a predetermined significance level. We use four loss functions, namely mean absolute deviation (MAD), mean squared error (MSE), mean absolute percentage error (MAPE), and polynomial function (poly), to compute the DM statistics, as shown in Table 11. Values marked with an underline indicate non-significance. In crude oil price prediction, all results are negative and significant at the 1% level, indicating that using ES-CF-BiAt for point prediction of crude oil price is significantly superior to other models. However, in the prediction of crude oil volatility, when setting the loss functions to MAD and MAPE, we observe two non-significant values (-2.52 and 2.07) and three positive values (3.53, 19.73, and 2.07), while the majority of results are negative and significant at the 1% level. In summary, our proposed ESMD-CFastICA-BiLSTM-Attention model demonstrates advantages in data decomposition, reconstruction, and deep learning model selection, with robust results.

Interval prediction quantifies the variations in prediction results caused by uncertain factors, where the original value falls within a prediction interval determined by the upper and lower bounds at a given significance level. Specifically, given a significance level 2α , the estimated value of the actual observation X_t (in our study, crude oil price and volatility) is considered as the expectation for the future. The probability that the estimated value falls within its upper bound U and lower bound L satisfies $P(L \leq X_t \leq U) = P(L \leq X_t \leq U | E(X_t) = \hat{x}) \times P(E(X_t) = \hat{x})$. Further, assuming that the distribution function f

Table 11
The results of DM test.

Model	Price				Volatility			
	MAD	MSE	MAPE	Poly	MAD	MSE	MAPE	Poly
EE-CF-BiAt	-47.66	-17.50	-48.35	-17.50	-11.25	-3.69	-24.34	-3.69
E-CF-BiAt	-38.75	-14.59	-43.97	-14.59	-19.42	-3.47	-47.36	-3.47
ES-BiAt	-38.61	-20.81	-45.94	-20.81	-2.52	-3.72	2.07	-3.72
S-BiAt	-29.23	-9.94	-26.18	-9.94	-4.94	-3.44	-4.41	-3.44
ES-CF-BiLSTM	-28.30	-8.36	-35.29	-8.36	-9.93	-3.79	-27.86	-3.79
ES-CF-LSTM	-29.59	-11.50	-20.17	-11.50	-47.79	-10.17	-67.36	-10.17
ES-CF-SVR	-108.13	-57.53	-114.24	-57.53	-205.06	-33.75	-143.06	-33.75
ES-CF-RF	-32.13	-28.75	-26.85	-28.75	3.53	-4.27	19.73	-4.27

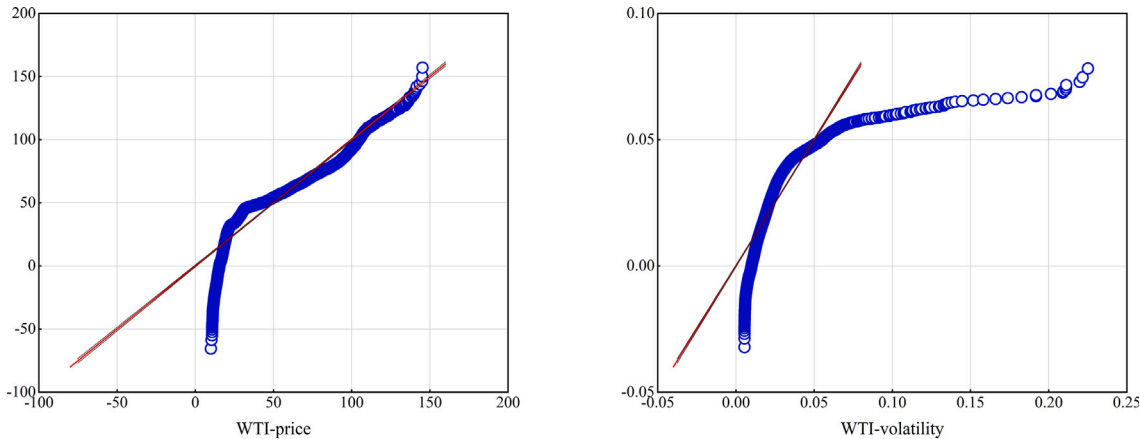


Fig. 10. Q-Q plot of crude oil price and volatility.

of the prediction values is similar to that of the real values, and the variance of the unknown points is the same as the historical variance S^2 . We set $\int z f(z | \theta) dz = \hat{x}$ and $\int (z - E(z))^2 f(z | \theta) dz = S^2$, and then calculate the upper and lower bounds of the prediction interval using $\{(\hat{L}, \hat{U}) | \hat{L} \leq \hat{x}_t \leq \hat{U}, \int_{\hat{L}}^{\hat{U}} f(z | \theta) dz + \int_{\hat{L}}^{\hat{x}} f(z | \theta) dz = 1 - 2\alpha\}$. Therefore, it is necessary to determine the distribution characteristics of the true values.

First, we use QQ plots to test whether crude oil price and volatility follow a normal distribution. By ordering the test data from smallest to largest, calculating the corresponding cumulative percentiles, and comparing them with a normal distribution, we examine the distribution of crude oil price and volatility. Fig. 10 shows the QQ plots for crude oil price and volatility sequences, where the x -axis represents the quantiles of the normal distribution, the y -axis represents the sample quantiles, the red line represents the normal distribution, and the blue dots represent the test data. The results indicate that the blue dots deviate from the red line, suggesting that crude oil price and volatility do not follow a normal distribution. Additionally, we use the Jarque-Bera, Anderson-Darling, and Kolmogorov-Smirnov tests to examine crude oil price and volatility, finding that at the 1% significance level, the hypothesis of a normal distribution for crude oil price and volatility is rejected.

Next, we constructed histograms for the distribution of crude oil price and volatility. In Fig. 11, the histograms for both crude oil price and volatility are skewed to the left, with the peak of the data also on the left. Therefore, we consider that the crude oil price and volatility follow a log-normal distribution rather than a normal distribution.

We further elaborate on the interval prediction process and evaluate the results from two perspectives, after determining the distribution characteristics of crude oil price and volatility. One aspect is to assess the impact of different significance levels on the interval prediction results of the ESMD-CFastICA-BiLSTM-Attention model. The other aspect is to validate the advantage of our proposed new composite model in predicting crude oil price and volatility by comparing the interval prediction results of different models. Having established that the true

Table 12
The interval prediction at different significance levels.

α	Price			Volatility		
	PICP	PIAW	AWD	PICP	PIAW	AWD
0.1	0.9659	11.2426	0.0142	0.9322	0.0073	0.0306
0.15	0.9543	9.8360	0.0190	0.9165	0.0064	0.0408
0.2	0.9411	8.7547	0.0245	0.9012	0.0057	0.0518
0.25	0.9228	7.8571	0.0392	0.8836	0.0051	0.0642
0.3	0.9033	7.0781	0.0392	0.8685	0.0046	0.0784
0.35	0.8785	6.3820	0.0494	0.8493	0.0042	0.0950
0.4	0.8497	5.7467	0.0624	0.8220	0.0037	0.1148

values of crude oil price and volatility follow a log-normal distribution, we calculate the mean and variance using the full sample data from April 5, 1983, to December 12, 2023. We set significance levels α at 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, and 0.4, and calculate the corresponding upper and lower bounds. The results are evaluated using the three indicators PICP, PIAW, and AWD from Table 4, as shown in Table 12. We observe that in the prediction of crude oil price and volatility, as the significance level increases, the values of PICP and PIAW gradually decrease, while AWD gradually increases. This indicates that with the increase in significance level, the prediction interval width (PIAW) decreases, leading to a smaller proportion of true values falling within the interval and a larger proportion falling outside the interval. This is reflected by a smaller PICP and a larger AWD.

Fig. 12 visualizes the interval prediction results for crude oil price and volatility at a significance level of 0.4. The prediction intervals are represented by gray shaded areas, while the blue and red solid lines represent the real values of WTI crude oil price and volatility, respectively. From the figure, we can observe that although the PICP is relatively low and the AWD is high at a significance level of 0.4, the upper and lower bounds of the prediction intervals follow a similar trend to the real values, and most of the actual values fall within

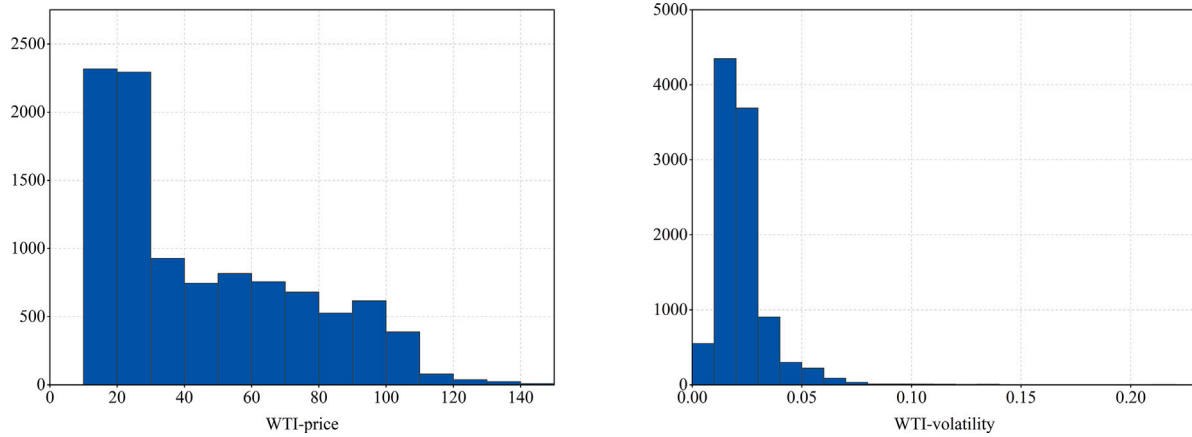


Fig. 11. The histogram of crude oil price and volatility.

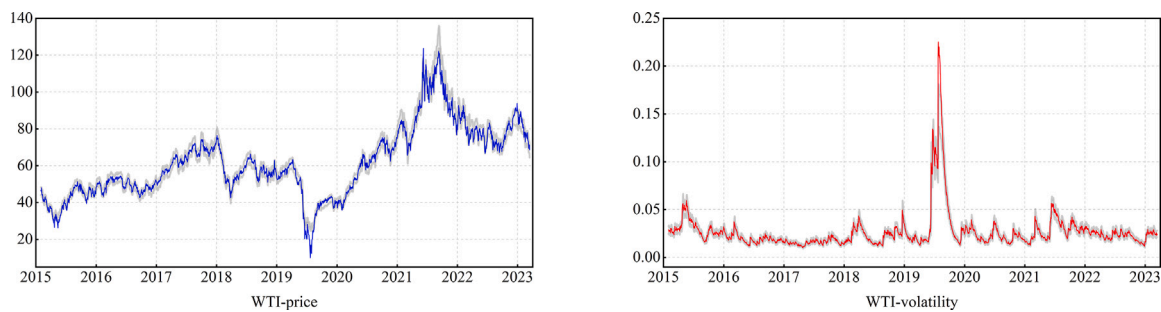


Fig. 12. The prediction intervals of crude oil price and volatility via ES-CF-BiAt.

the prediction intervals. This indicates that the ESMD-FastICA-BiLSTM-Attention model is capable of performing interval prediction for crude oil price and volatility.

We evaluate the interval prediction results of different models using three metrics: PICP, PIAW, and AWD. When PIAW is constant, a larger PICP and a smaller AWD indicate better interval prediction results. For clarity, we set the significance level of the interval prediction for ESMD-CFastICA-BiLSTM-Attention to 0.4, calculated the mean and variance of the test set, and combined the point prediction results of the test set to calculate the corresponding prediction intervals. The results show that for crude oil price, PICP = 0.9134, PIAW = 5.7467, AWD = 0.0551, and for crude oil volatility, PICP = 0.8220, PIAW = 0.0037, AWD = 0.1148. Further, we fixed the PIAW for interval prediction of crude oil price and volatility, used other comparative models for interval prediction, and compared the results with ES-CF-BiAt. Table 13 shows the interval prediction results for all models, with the best values for each metric in bold. For crude oil price interval prediction, the PIAW is set to 5.7467, and ES-CF-BiAt achieves the best PICP, demonstrating its superiority in crude oil price interval prediction. Similarly, in crude oil volatility prediction, we fixed the PIAW of the prediction interval to 0.0037. Compared to other models, ES-CF-BiAt has the highest PICP of 0.8220 and the smallest AWD of 0.1148. Therefore, the interval prediction results of different models indicate that the combination of ESMD decomposition, CFastICA reconstruction, and BiLSTM-Attention can improve the interval prediction accuracy of crude oil price and volatility. Our proposed model also demonstrates an advantage in interval prediction for crude oil price and volatility.

4.3. Robustness test

Our proposed ESMD-FastICA-BiLSTM-Attention predictive model is capable of forecasting not only crude oil prices but also crude oil volatility, indicating that this novel integrated model possesses a

Table 13

The interval prediction results with different models.

Model	Price			Volatility		
	PICP	PIAW	AWD	PICP	PIAW	AWD
ES-CF-BiAt	0.9134	5.7467	0.0551	0.8220	0.0037	0.1148
EE-CF-BiAt	0.7034	5.7467	0.0793	0.5237	0.0037	0.2970
E-CF-BiAt	0.8722	5.7467	0.0618	0.6226	0.0037	0.1854
ES-BiAt	0.4846	5.7467	0.2498	0.7107	0.0037	0.2202
S-BiAt	0.8277	5.7467	0.0515	0.6862	0.0037	0.7756
ES-CF-BiLSTM	0.8894	5.7467	0.0481	0.6153	0.0037	0.4175
ES-CF-LSTM	0.8228	5.7467	0.0454	0.1674	0.0037	2.8895
ES-CF-SVR	0.1219	5.7467	2.4179	0.0416	0.0037	1.9521
ES-CF-RF	0.1850	5.7467	1.0902	0.5335	0.0037	0.4042

certain level of stability. To further test the generalization ability of ESMD-FastICA-BiLSTM-Attention, we conduct robustness checks using the China crude oil futures price index(INE crude oil price) from the Shanghai International Energy Exchange. Fig. 13 illustrates the trend of INE crude oil price from March 20, 2018, to December 8, 2023. We observe that the variation pattern of INE crude oil prices is similar to the trend of WTI during the same period, showing an overall upward trend. The volatility increased in 2022, influenced by the Russia-Ukraine war.

First, we decompose the INE crude oil price into 9 intrinsic modes using ESMD. Then, we reconstruct these intrinsic modes and extract independent components through CFastICA. Finally, we use these independent components as inputs for BiLSTM-Attention to perform both point prediction and interval prediction of INE crude oil price. Additionally, we use other composite models from Table 6 to predict INE crude oil price, with all results presented in Table 14. In point prediction, the three metrics – MAE, RMSE, and R^2 – all show that ES-CF-BiAt achieves the best prediction results for INE crude oil price. This demonstrates that ESMD decomposition, CFastICA reconstruction, and



Fig. 13. The time series of INE crude oil price.

Table 14

The point and interval prediction results of INE price.

Model	Point prediction				Interval prediction		
	MAE	RMSE	R^2	D_{stat}	PICP	PIAW	AWD
ES-CF-BiAt	0.0457	0.0599	0.9128	0.5379	0.6079	27.7188	0.1387
EE-CF-BiAt	0.0622	0.0822	0.8358	0.7076	0.4928	27.7188	0.2457
E-CF-BiAt	0.0881	0.1019	0.7476	0.5451	0.2626	27.7188	0.4574
ES-BiAt	0.0649	0.0854	0.8229	0.7184	0.4928	27.7188	0.2603
S-BiAt	0.0770	0.0914	0.7974	0.5560	0.3237	27.7188	0.3627
ES-CF-BiLSTM	0.1766	0.1990	0.0386	0.5371	0.1259	27.7188	1.3419
ES-CF-LSTM	0.1496	0.1687	0.3091	0.5307	0.1151	27.7188	1.0481
ES-CF-SVR	0.1691	0.1872	0.1493	0.5451	0.0612	27.7188	1.2011
ES-CF-RF	0.3538	0.3862	-2.6220	0.4765	0.0432	27.7188	3.2472

Table 15

Effectiveness and significance evaluations of INE price prediction results.

Model	Effectiveness evaluation		Significance evaluation			
	I_{MAE} (%)	I_{RMSE} (%)	MAD	MSE	MAPE	Ploy
EE-CF-BiAt	36.22	37.22	-5.87	-5.56	-5.20	-5.56
E-CF-BiAt	93.05	70.12	-12.14	-10.81	-13.62	-10.81
ES-BiAt	42.22	42.50	-6.39	-5.90	-5.63	-5.90
S-BiAt	68.60	52.44	-10.01	-8.68	-11.28	-8.68
ES-CF-BiLSTM	286.79	232.03	-23.39	-18.33	-22.29	-18.33
ES-CF-LSTM	227.64	181.48	-25.35	-18.25	-23.34	-18.24
ES-CF-SVR	270.33	212.34	-24.04	-17.36	-24.31	-17.35
ES-CF-RF	674.92	544.48	-32.53	-20.53	-32.54	-20.53

BiLSTM-Attention all play important roles in improving the accuracy of point prediction for crude oil prices. For interval prediction, we assume that INE crude oil price follows a log-normal distribution and set the significance level to 0.8, resulting in a PIAW of 27.7188. Based on this, we fix the PIAW for other models at the same value to calculate PICP and AWD. We find that under the same bandwidth, ES-CF-BiAt has the highest number of real values falling within the prediction interval and the fewest real values outside the interval. This indicates that ES-CF-BiAt can also enhance the accuracy of interval prediction for crude oil prices. Therefore, the data preprocessing and deep learning methods we selected can increase the prediction advantages.

Furthermore, we evaluate the effectiveness and significance of the prediction results for all models. The evaluation results are shown in Table 15. For ES-CF-BiAt, the improvement percentages in MAE and RMSE compared to other models are all above 30%. The DM statistics calculated using the four evaluation functions are significantly negative at the 1% level. This indicates that ESMD-FastICA-BiLSTM-Attention can improve the prediction accuracy of INE crude oil price and is statistically significant. Therefore, our proposed novel integrated model can be used for forecasting in the crude oil market, and the results are robust.

5. Conclusion

Effective data analysis and prediction methods play a crucial role in trend forecasting in the crude oil market. However, crude oil price and volatility exhibit complex data features, posing significant challenges to enhancing prediction accuracy. We combine signal processing methods and deep learning models to propose a novel integrated model, offering new perspectives on predicting market trends based on existing research. Firstly, crude oil volatility is computed based on the

crude oil price sequence. Extreme-point symmetric empirical mode decomposition is employed to decompose crude oil price and volatility, obtaining multiple subseries with different frequencies. Secondly, K-means clustering and fast independent component analysis are used to reconstruct the decomposed subseries and extract independent components, ensuring improved prediction efficiency while fully exploring data information. Finally, these independent components are used as inputs, combined with bidirectional long short-term memory network and attention mechanism models to achieve point prediction and interval prediction for crude oil price and volatility.

The conclusions of this study can be summarized in three main aspects: (1) The decomposition and reconstruction of data play a crucial role in extracting historical information on crude oil prices and volatility. Compared to decomposition alone, CFastICA can compensate for the limitations of ESMD decomposition, significantly enhancing the accuracy and efficiency of point and interval predictions. (2) Compared to single prediction models, deep learning ensemble models perform better in point and interval predictions for crude oil price and volatility. The attention mechanism endows the prediction model with discernment, enhancing the performance of point and interval predictions. (3) Different significance levels affect the average width of interval predictions for crude oil price and volatility. At a determined average bandwidth, the proposed model in this paper demonstrates the best interval prediction results for crude oil price and volatility.

While our study has achieved certain results, there are still two main limitations. Firstly, there are limitations in the research methods. The data preprocessing methods we used can be further expanded, especially the reconstruction methods, as we did not compare multiple reconstruction methods and select the optimal results. Although the selected deep learning models performed best in point and interval predictions, we did not consider cross-domain models and the interpretability of the models during the research process. Secondly, there

are limitations in the research content. Our predictions of crude oil prices and volatility are based only on the historical time series of crude oil, without considering the multidimensional influencing factors of crude oil prices and volatility. Additionally, the evaluation metric system for prediction results needs further improvement. Therefore, as a research extension, we plan to use different data preprocessing methods and deep learning methods to enhance prediction accuracy while improving model interpretability. At the same time, we will incorporate research methods from other fields, such as trajectory similarity, into the crude oil market. We will consider the complex influencing factors of the crude oil market, add sensitivity analysis of the models, and improve the effectiveness and significance evaluation metrics system, conducting research on predictions for multiple crude oil markets.

CRedit authorship contribution statement

Zisheng Ouyang: Writing – review & editing, Project administration, Funding acquisition, Conceptualization,. **Min Lu:** Writing – original draft, Software, Methodology, Data curation. **Zhongzhe Ouyang:** Writing – review & editing, Methodology, Data curation. **Xuewei Zhou:** Writing – review & editing, Validation, Supervision. **Ren Wang:** Writing – review & editing, Visualization, Software.

Acknowledgments

This research was supported by the National Social Science Fund of China (Grant no. 23BTJ043) and Hunan Key Laboratory of Macroeconomic Big Data Mining and its Application, China.

References

- Abdollahi, H., Ebrahimi, S.B., 2020. A new hybrid model for forecasting Brent crude oil price. *Energy* 200, 117520.
- Badeeb, R.A., Szulczyk, K.R., Lean, H.H., 2021. Asymmetries in the effect of oil rent shocks on economic growth: A sectoral analysis from the perspective of the oil curse. *Resour. Policy* 74, 102326.
- Bucci, A., 2020. Realized volatility forecasting with neural networks. *J. Financ. Econom.* 18 (3), 502–531.
- Busari, G.A., Lim, D.H., 2021. Crude oil price prediction: A comparison between AdaBoost-LSTM and AdaBoost-GRU for improving forecasting performance. *Comput. Chem. Eng.* 155, 107513.
- Chen, Y.C., Huang, W.C., 2021. Constructing a stock-price forecast CNN model with gold and crude oil indicators. *Appl. Soft Comput.* 112, 107760.
- Chen, K., Wen, H., 2008. A study on quick selective bagging trees ensemble algorithm based on K-means cluster technology. *J. Stat. Inf.* (09), 23–27.
- Cheong, C.W., 2009. Modeling and forecasting crude oil markets using ARCH-type models. *Energy Policy* 37 (6), 2346–2355.
- Chiroma, H., Abdulkareem, S., Herawan, T., 2015. Evolutionary neural network model for West Texas Intermediate crude oil price prediction. *Appl. Energy* 142, 266–273.
- Drachal, K., 2021. Forecasting crude oil real prices with averaging time-varying VAR models. *Resour. Policy* 74, 102244.
- Fang, T., Zheng, C., Wang, D., 2023. Forecasting the crude oil prices with an EMD-ISBM-FNN model. *Energy* 263, 125407.
- Guha, B., Bandyopadhyay, G., 2016. Gold price forecasting using ARIMA model. *J. Adv. Manag. Sci.* 4 (2).
- Guo, L., Huang, X., Li, Y., Li, H., 2023. Forecasting crude oil futures price using machine learning methods: Evidence from China. *Energy Econ.* 127, 107089.
- Guo, J., Zhao, Z., Sun, J., Sun, S., 2022. Multi-perspective crude oil price forecasting with a new decomposition-ensemble framework. *Resour. Policy* 77, 102737.
- Hasan, M.M., Asad, S., Wong, J.B., 2022. Oil price uncertainty and corporate debt maturity structure. *Finance Res. Lett.* 46, 102278.
- Hyvärinen, A., Oja, E., 2000. Independent component analysis: algorithms and applications. *Neural Netw.* 13 (4–5), 411–430.
- Ilyas, M., Khan, A., Nadeem, M., Suleman, M.T., 2021. Economic policy uncertainty, oil price shocks and corporate investment: Evidence from the oil industry. *Energy Econ.* 97, 105193.
- Jiang, H., Hu, W., Xiao, L., Dong, Y., 2022. A decomposition ensemble based deep learning approach for crude oil price forecasting. *Resour. Policy* 78, 102855.
- Jiang, Y., Tian, G., Mo, B., 2020. Spillover and quantile linkage between oil price shocks and stock returns: new evidence from G7 countries. *Financ. Innov.* 6, 1–26.
- Jianwei, E., He, K., Liu, H., Ji, Q., 2023. A novel separation-ensemble analyzing and forecasting method for the gold price forecasting based on RLS-type independent component analysis. *Expert Syst. Appl.* 120852.
- Kilian, L., Vigfusson, R.J., 2013. Do oil prices help forecast US real GDP? The role of nonlinearities and asymmetries. *J. Bus. Econom. Statist.* 31 (1), 78–93.
- Li, Y., Jiang, S., Li, X., Wang, S., 2021a. The role of news sentiment in oil futures returns and volatility forecasting: Data-decomposition based deep learning approach. *Energy Econ.* 95, 105140.
- Li, M., Li, Y., 2023. Research on crude oil price forecasting based on computational intelligence. *Data Sci. Finance Econ.* 3 (3), 251–266.
- Li, T., Qian, Z., Deng, W., Zhang, D., Lu, H., Wang, S., 2021b. Forecasting crude oil prices based on variational mode decomposition and random sparse Bayesian learning. *Appl. Soft Comput.* 113, 108032.
- Li, G., Yin, S., Yang, H., 2022. A novel crude oil prices forecasting model based on secondary decomposition. *Energy* 257, 124684.
- Lin, Y., Chen, K., Zhang, X., Tan, B., Lu, Q., 2022. Forecasting crude oil futures prices using BiLSTM-Attention-CNN model with Wavelet transform. *Appl. Soft Comput.* 130, 109723.
- Liu, J., Huang, X., Li, Q., Chen, Z., Liu, G., Tai, Y., 2023. Hourly stepwise forecasting for solar irradiance using integrated hybrid models CNN-LSTM-MLP combined with error correction and VMD. *Energy Convers. Manage.* 280, 116804.
- Liu, W., Liu, W.D., Gu, J., 2020a. Forecasting oil production using ensemble empirical model decomposition based Long Short-Term Memory neural network. *J. Pet. Sci. Eng.* 189, 107013.
- Liu, Y., Yang, C., Huang, K., Gui, W., 2020b. Non-ferrous metals price forecasting based on variational mode decomposition and LSTM network. *Knowl.-Based Syst.* 188, 105006.
- Lu, W., Li, J., Wang, J., Qin, L., 2021. A CNN-BiLSTM-AM method for stock price prediction. *Neural Comput. Appl.* 33, 4741–4753.
- Luo, C., Liu, L., Wang, D., 2021. Multiscale financial risk contagion between international stock markets: Evidence from EMD-Copula-CoVaR analysis. *N. Am. J. Econ. Finance* 58, 101512.
- Luo, C., Qu, Y., Su, Y., Dong, L., 2024. Risk spillover from international crude oil markets to China's financial markets: Evidence from extreme events and US monetary policy. *N. Am. J. Econ. Finance* 70, 102041.
- Ma, C., Dai, G., Zhou, J., 2021. Short-term traffic flow prediction for urban road sections based on time series analysis and LSTM-BiLSTM method. *IEEE Trans. Intell. Transp. Syst.* 23 (6), 5615–5624.
- Niu, Z., Demirel, R., Suleman, M.T., Zhang, H., Zhu, X., 2023a. Do industries predict stock market volatility? Evidence from machine learning models. *J. Int. Financ. Mark. Inst. Money* 101903.
- Niu, Z., Wang, C., Zhang, H., 2023b. Forecasting stock market volatility with various geopolitical risks categories: New evidence from machine learning models. *Int. Rev. Financ. Anal.* 89, 102738.
- Ouyang, Z., Lu, M., Lai, Y., 2023. Forecasting stock index return and volatility based on GAVMD-Carbon-BiLSTM: How important is carbon emission trading? *Energy Econ.* 128, 107134.
- Ouyang, Z., Lu, M., Zhou, X., 2022. A study of financial risk spillover and early warning in China based on TVP-VAR-LSTM model. *J. Stat. Inf.* 37 (10), 53–64.
- Ouyang, Z., Yang, X., Lai, Y., 2021. Systemic financial risk early warning of financial market in China using Attention-LSTM model. *N. Am. J. Econ. Finance* 56, 101383.
- Ozdemir, A.C., Buluş, K., Zor, K., 2022. Medium-to long-term nickel price forecasting using LSTM and GRU networks. *Resour. Policy* 78, 102906.
- Sinaga, K.P., Yang, M.-S., 2020. Unsupervised K-means clustering algorithm. *IEEE Access* 8, 80716–80727.
- Urolagin, S., Sharma, N., Datta, T.K., 2021. A combined architecture of multivariate LSTM with Mahalanobis and Z-Score transformations for oil price forecasting. *Energy* 231, 120963.
- Wang, L., An, H., Liu, X., Huang, X., 2016. Selecting dynamic moving average trading rules in the crude oil futures market using a genetic approach. *Appl. Energy* 162, 1608–1618.
- Wang, J., Cao, J., Yuan, S., Cheng, M., 2021. Short-term forecasting of natural gas prices by using a novel hybrid method based on a combination of the CEEMDAN-SE and the PSO-ALS-optimized GRU network. *Energy* 233, 121082.
- Wang, J.L., Li, Z.J., 2013. Extreme-point symmetric mode decomposition method for data analysis. *Adv. Adapt. Data Anal.* 5 (03), 1350015.
- Wang, J., Wang, Z., Li, J., Wu, J., 2018. Multilevel wavelet decomposition network for interpretable time series analysis. In: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. pp. 2437–2446.
- Wen, D., Liu, L., Wang, Y., Zhang, Y., 2022. Forecasting crude oil market returns: Enhanced moving average technical indicators. *Resour. Policy* 76, 102570.
- Wu, Y.X., Wu, Q.-B., Zhu, J.Q., 2019. Improved EEMD-based crude oil price forecasting using LSTM networks. *Phys. A* 516, 114–124.
- Xu, K., Niu, H., 2023. Denoising or distortion: Does decomposition-reconstruction modeling paradigm provide a reliable prediction for crude oil price time series? *Energy Econ.* 128, 107129.
- Yang, D.X., Wu, B.B., Tong, J.Y., 2021. Dynamics and causality of oil price shocks on commodities: Quantile-on-quantile and causality-in-quantiles methods. *Resour. Policy* 74, 102246.
- Yu, Y., Guo, S., Chang, X., 2022. Oil prices volatility and economic performance during COVID-19 and financial crises of 2007–2008. *Resour. Policy* 75, 102531.
- Zhang, X., Cheng, S., Zhang, Y., Wang, J., Wang, S., 2024. An attention-PCA based forecast combination approach to crude oil price. *Expert Syst. Appl.* 240, 122463.

- Zhang, Y., Huang, Y., Wang, X., 2023a. Impact of economic policy uncertainty, oil prices, and technological innovations on natural resources footprint in BRICS economies. *Resour. Policy* 86, 104082.
- Zhang, S., Luo, J., Wang, S., Liu, F., 2023b. Oil price forecasting: A hybrid GRU neural network based on decomposition–reconstruction methods. *Expert Syst. Appl.* 218, 119617.
- Zhu, B., Han, D., Wang, P., Wu, Z., Zhang, T., Wei, Y.M., 2017. Forecasting carbon price using empirical mode decomposition and evolutionary least squares support vector regression. *Appl. Energy* 191, 521–530.